

Xiongxin Yang

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Education

University of California, Santa Barbara

Sept 2025 – Present

Ph.D. Student in Computer Science

◦ Advisor: Eric Vigoda

Northeast Normal University

Sept 2020 – Jun 2024

B.S. in Computer Science and Technology

◦ Advisor: Zhiguo Fu

Employment

The University of Hong Kong

May 2025 – Jul 2025

Research Assistant hosted by Weiming Feng

New York University Shanghai

Oct 2024 – Jan 2025

Research Associate hosted by Jie Xue

Visits & Internships

The University of Edinburgh

Jul 2026

Visiting Student hosted by Heng Guo

The University of Hong Kong

Mar 2025

Visitor hosted by Weiming Feng

Nanjing University

Apr 2023

Visiting Student hosted by Yitong Yin

Selected Awards

Presidential Scholarship (Northeast Normal University)

2022 & 2024

Ranked 1st in the major

National Scholarship

2021

Top national scholarship in mainland China (top 0.2% nationwide)

Publications & Preprints

[10] Fast FPRAS for the Permanent

Xiaoyu Chen, Heng Guo, Eric Vigoda, **X. Yang**

[arXiv: 2609.20717](#)

[9] Optimal Simulated Annealing for Partition Function Estimation

Heng Guo, Hongyang Liu, **X. Yang**, Yitong Yin, Yiyao Zhang

[arXiv: 2609.20337](#)

[8] Faster FPRAS for the Permanent via Restricted Poincaré Inequalities and Coupled Flows

Xiaoyu Chen, Eric Vigoda, **X. Yang**

[arXiv: 2608.26599](#)

- [7] 0/1-Polytopes with Exponentially Small Edge Expansion
X. Yang
[arXiv: 2608.01870](#)
- [6] Learning AC^0 under Locally Sampleable Graphical Models
 Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
[arXiv: 2607.08303](#)
- [5] Near-Optimal Dynamic Data Structures for Maximum Depth and Klee's Measure of Boxes
 Sujoy Bhore, Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
 53rd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2026**)
- [4] Learning CNF formulas from uniform random solutions in the local lemma regime
 Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
 58th Annual ACM Symposium on Theory of Computing (**STOC 2026**)
- [3] Dynamic maximum depth of geometric objects
 Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
 41st International Symposium on Computational Geometry (**SoCG 2025**)
- [2] Spectral Independence Beyond Total Influence on Trees and Related Graphs
 Xiaoyu Chen, **X. Yang**, Yitong Yin, Xinyuan Zhang
 2025 ACM-SIAM Symposium on Discrete Algorithms (**SODA 2025**)
- [1] Beyond windability: Approximability of the four-vertex model
 Tianyu Liu, **X. Yang**
Theoretical Computer Science, Volume 995 (2024), Article 114491

Talks & Posters

- Near-Optimal Dynamic Maximum Depth and Klee's Measure of Boxes
 - Jul 8, 2026 @ ICALP 2026: Royal Holloway, University of London, UK
- Learning CNF formulas from uniform random solutions in the local lemma regime
 - Jun 22, 2026 @ STOC 2026 Online Poster Session
- Dynamic Maximum Depth of Geometric Objects
 - Jun 25, 2025 @ SoCG 2025: Kanazawa, Ishikawa Prefecture, Japan
 - May 31, 2025 @ AAAC 2025: The City University of Hong Kong, Hong Kong SAR, China
- A Glimpse into the Theory World
 - Apr 11, 2025 @ Information Science Academic Forum: Northeast Normal University, Changchun, China
- Approximability of the Four-Vertex Model
 - Oct 21, 2024 @ NYUSH Theory Lunch: New York University Shanghai, Shanghai, China

Services

Conference Reviewing: WWW (2026), SoCG (2025), SODA (2027), SOSA (2027)